Karnataka

Chemistry 2017

Instruction:

(1) The question paper has four parts. All parts are compulsory
(2) Part-A carries 10 marks. Each question carries two marks.
Part-B carries 10 marks. Each question carries two marks.
Part –C carries 15 marks. Each question carries three marks.
Part-D carries 35 marks. Each question carries five marks.
(3) Write balanced chemical equations and draw diagrams wherever necessary.
(4) Use log tables and simple calculator if necessary.

PART-A

I. Answer all the questions. Each questions carries one mark. (Answer each question in be word or in one sentence)

1. How molarity does varies with temperature?

Sol: Molarity decrease with increase in temperature.

2. 10mL of liquid 'A' mixed with 10mL of liquid 'B', the volume of the resultant solution id 19.9mL. What type of deviation expected from Rault's law?

Sol: Negative

3. Write the mathematical expression for limiting molar conductivity of sodium chloride $[NaCl]$

Sol: $\wedge^0_{(NaCl)} = \lambda^0_{Na} + \lambda^0_{Cl^-}$

4. Define collision property.

Sol: The number of collisions per second per unit volume of the reaction mixture.

5. Name the adsorbent used to removal of coloring matter from solution.

Sol: Animal charcoal/ Activated charcoal/ Charcoal/Activated carbon.

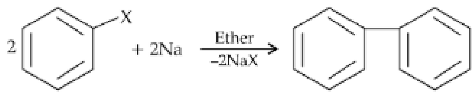
6. Give an example of metal purified by Mond process.

Sol: Nickel

7. Which noble gas is most abundant in atmospheric dry air?

Sol: Argon

8. What is the name of the following reaction?



Sol: Fittig's reaction.

9. Formaldehyde $HCHO$ undergoes Cannizzaro reaction: Give reason.

Sol: It does not contain α - hydrogen atom.

10. Deficiency of which vitamin causes the disease scurvy.

Sol: Vitamin-C

PART – B

II. Answer any five of the following. Each question carries 2 marks.

11. Give the difference between crystalline and amorphous solids with respect to shape and melting point.

Sol:

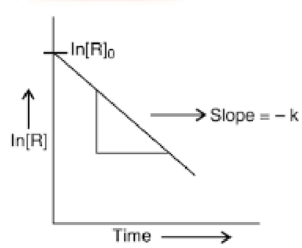
	Property	**Crystalline**	**Amorphous solid**
1	Shape	Definite geometrical shape	No definite geometrical shape/ Irregular shape
2	Melting point	Sharp (Definite) melting point.	No sharp (Definite) melting point/Melt over range of temperature.

12. Write the cathodic and anodic cell reactions of Hydrogen-Oxygen fuel cell.

Sol:

$$Cathode:\ O_2 + 2H_2 + 4e^- \rightarrow 4OH^-$$
$$Anode: 2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$$

13. From the following graph, identify order of reaction and mention the unit of its rate constant.



Sol:

$Order$: first/One/1^{st}

$Unit: Time^{-1}\left(\sec^{-1} or \min^{-1} \text{ or hrs}^{-1}\right)$

Lanthanide contraction: The steady or gradual decrease in atomic and ionic radii from lanthanum lutetium is called lanthanide contraction.

It occurs due to

- Imperfect shielding of one electron by another in the same sub-shell.
- Poor shielding effect of 4f electrons.

14. What is lanthanide contraction? Mention the cause for it.

Sol: Lanthanide contraction:

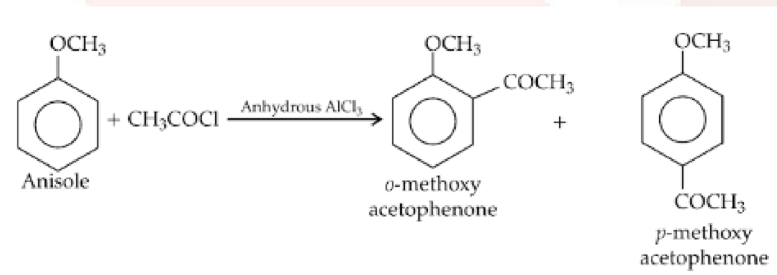
The steady or gradual decrease in atomic and ionic radii from lanthanum to leutetium is called lanthanoid contraction.

It occurs due to

- Imperfect shilding of one electron by another in the same sub-shell (same set of orbital's).
- Poor shielding effect of 4f electron.

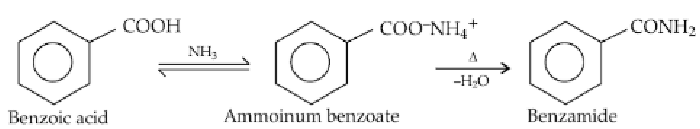
15. How anisole reacts with acetyl chloride $[CH_3COCl]$ in the presence of anhydrous $AlCl_3$ Write the equation for the reaction.

Sol: When anisole reacts with actyle chloride in the presence of anhydrous $AlCl_3$, p-gives a mixture



16. What is the action of ammonia $[NH_3]$ on benzoic acid? Write equation.

Sol: When ammonia reacts with benzoic acid, ammonium benzoate is formed which on heating gives Benzamide.



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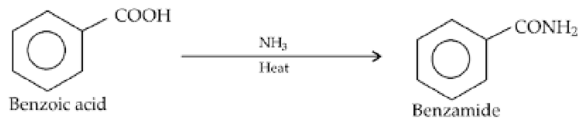


17. What are anionic detergents? Give an example.

Sol: Self-explanatory equation

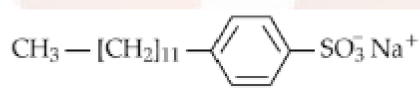
(i) Aspririn (2-acetoxy benzoic acid) OR Paracetamol (OR 4-acetamidophenol) Or Diclofenac sodium OR Diclofenac ppotassium OR Any suitable example.

(ii) Furacine OR Soframycine OR Dettol (ORchloroxylenol andterpineol) OR Bithionol OR Tincture of Iodine (2-3% iodine solution in alcohol) OR IOdoformoric acid in dilute aqueous solution OR 0.2% phenol OR hydrogen peroxide OR Hexachloriphene OR Amyl metacresol(5-methyl-2pentyl phenol).

18. What are anionic detergents? Give an example.

Sol: Sodium salts of sulphonated long chain alcohols or hydrocarbons. Detergents whose cleaning action is due to anion present in it.

Example,

Sodium dodecyl benzene sulphonate Or 

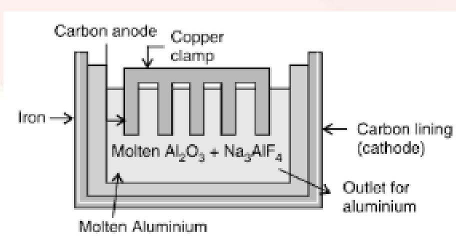
PART – C

(II). Answer any five of the following. Each question carries 3 marks.

19. During the extraction of aluminum by Hall- Heroult process.

(i) Write neat labelled diagram of electrolyte cell.

Sol:



(i) Write over all cell reaction.

Sol:

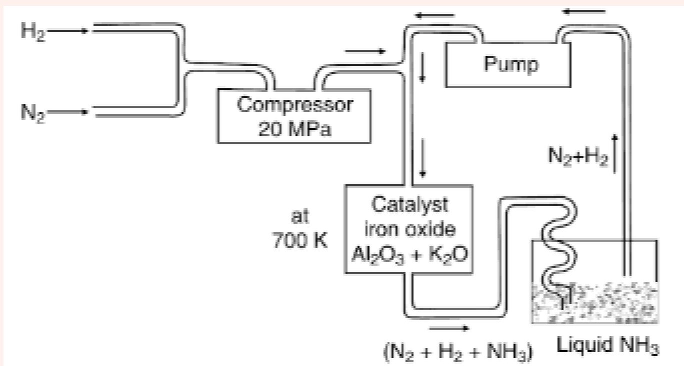
$2Al_2O_3 + 3C \rightarrow 4Al + 3CO_2$

(ii) At which electrode oxygen gas libreted.

Sol: Anode OR positive electrode

20. In the manufacture of ammonia by Haber's process, write the flow chart and chemical equation optimum condition.

Sol:



$N_2 + 3H_2 \rightleftharpoons 2NH_3 \Delta h_F^0 = -46Kj / MOL^{-1}$

The most favorable conditions according to Le-Chatelier's principle.

(i) The temperature is about 770K
(ii) The pressure is about 200×10^5
(iii) Iron oxide catalyst with small amount K_2O / Al_2O_3

21. (i) Mention any two reasons for the anomalous behavior of oxygen.

Sol: Small size, High electro negativity, Absence of d- orbitals, High ionization enethelpy.

(ii) Write the balanced chemical equation for the action of concentrated sulphuric acid on copper.

Sol:

$Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$

22. (i) Complete the following equations.

(a) $2NaOH + Cl_2 \rightarrow$

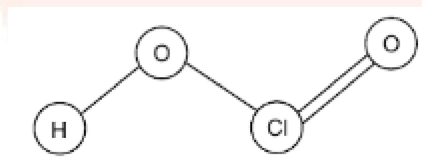
Sol: $Nacl + NaOCl + H_2O$

(b) $Cl_2 + 3F_2 \rightarrow \xrightarrow{573K}$

Sol: $2CIF_3$

(ii) Write the structure of chlorous acid $[HOClO]$.

Sol:



23. (i) Calculate the spin-only magnetic moment of Fe^{2+}.

Sol:

$$\mu = \sqrt{n(n+2)}$$
$$\mu = \sqrt{4(4+2)} = 4.90BM$$

(ii) Which element of 3d series exhibits maximum oxidation state?

Sol: Manganese

24. How is $\wedge_m^0 (Nacl) = \lambda_{Na}^0 + \lambda_{Cl^-}^0$ **is** MnO_2 ? **Write equation.**

Sol: Step1: The finely powdered MnO_2 is fused with KOH in the presence excess air and on oxidizing agent like KNO_3, to get green coloured potassium manganite.

$$2MnO_2 + 4KOH + O_2 \rightarrow 2K_2MNO_4 + 2H_2O$$

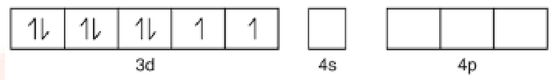
Step 2: Potassium manganite in acidic or neutral medium undergoes disproportionate to give potassium manganite.

$$3MNO_4^{2-} + 4h^+ \rightarrow 2MnO_4^- + MnO_2 + H_2O$$

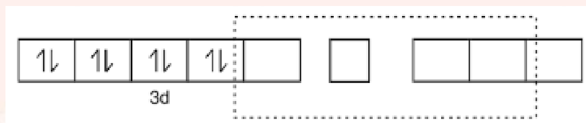
25. Explain hybridization, geometry and magnetic property of $\left[Ni\left(CN\right)_4\right]^{2-}$ ion using Valance Bond.

Sol:

Electronic configuration of $Ni^{2+} = \left[Ar\right]3d^8 4s^0$

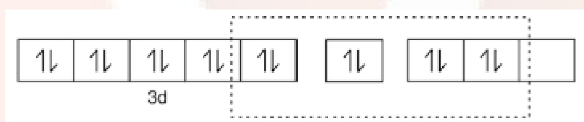


In the presence of strong ligand CN^- ion, the electronic configuration.



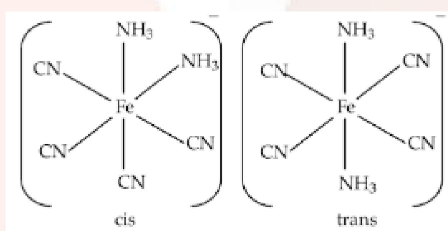
One 3d, one 4s and two 4p orbitals undergoes dsp^2 hybridization to give four equivalent vacant hybrid orbitals.

These four hybrid orbitals are occupied by four lone pairs of electrons from four CN^- ligands to form $\left[Ni\left(CN\right)_4^{2-}\right]$



26. (i) Write the cis and Trans isometric structure of $\left[Ni\left(CN\right)_4\right]^{2-}$ ion using Valance Bond.

Sol:



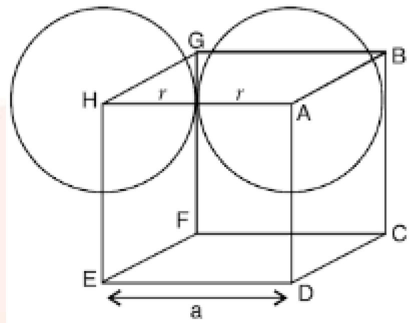
(ii) What is the co- ordination number of Fe in $\left[FeCl_2\left(en\right)_2\right]Cl$.

Sol: Six.

PART – D

27. (a) Calculate packing efficiency in simple cubic lattice.

Sol:



In a simple cubic unit cell, the number of particle is one i.e. considered as one sphere.

The volume of one particle (1 sphere) $=\frac{4}{3}\pi r^3$

Let the edge length of the cube $=a$

Radius of each particle =r

$a=2$r

The volume of the unit cell =

$a^3=(2r)^3=8r^3$

$packing$ effeciency= $\frac{volume \text{ occupied by 1 spheres}}{Totaal \text{ volume of the unit cell}}\times 100$

$Packing\ effincy\ \frac{\frac{4}{3}\pi r^3}{8r^3}\times 100=52.4\%$

(b) An element having atomic mas 63.1g/mol has face centred cubic unit cell with edge length 3.608 × 10^{-8} cm. Calculate the density of unit cell. [Given: N_A: 6.022 × 10^{23} atoms/mol]

Sol:

$d=\frac{ZM}{a^3N_A}$

$\Rightarrow d=\frac{4atoms\times 63.1g/mol}{\left(3.608\times 10^{-8}cm^3\right)\times\left(6.022\times 10^{23}\right)atom}$

$d=8.92gcm^{-3}$

28. (a)1.0g of non-electrolyte solute dissolved in 50g of benzene lowered the freezing point of benzene by 0.4K. Find the molar mass of the solute. [Given: Freezing point depression constant of benzene = 5.12 K.kgmol^{-1}]

Sol:

$$M_2 = \frac{k_f \times w_2 \times 1000}{\Delta T_f \times w_1}$$

$$M_2 = \frac{5.12kkgmol^{-1} \times 1.0g \times 100gkg^{-1}}{0.4 \times 50g}$$

$$M_2 = 256gmol^{-1}$$

(b) How solubility of a gas in liquid varies with (i) Temperature and (ii) pressure?

Sol: (i) Solubility decreases with increase in temperature or Solubility increase with decrease in temperature or solubility varies inversely with temperature.

(ii) Solubility increase with increase in pressure or Solubility decrease with decrease in pressure or solubility varies directly with pressure.

29. (a) The electrode potential for the Daniell cell given below is 1.1V.

Write overall cell reaction and calculate the standard Gibb's energy for the reaction. [F=96487 c/mol]

Sol:

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

$$\Delta G^0 = nFE^0_{cell}$$

$$\Delta G^0 = -2 \times 96487 \times 1.1 = -21227 J / mol$$

$$\Delta G^0 = -212 KJ / mol$$

(b) Mention any two factors which affects the conductivity of electrolytic solution.

(i) The nature of the electrolyte

(ii) Temperature

30. (a) Derive an integrated rate equation for rate constant of a zero order reaction.

Sol: Let us consider the following reaction which is of zero order.

The rate is given by:

$$R \to P$$
$$\frac{-d[R]}{dt} = k[R]^0$$
$$\frac{-d[R]}{dt} = k$$
$$-d[R] = kdt$$
$$d[R] = -kdt$$

On integration,

$$\int d[R] = -k\int dt$$
$$[R] = -kt + I$$

Where, I is integration constant

When t = 0

$[R] = [R_0]$ where R_0 is initial concentration of the reactant

$$[R_0] = -k \times 0 + I$$
$$[R_0] = I$$

Thus,

Equation (i) becomes,

$$[R] = -kt + [R_0]$$
$$kt = [R_0] - [R]$$
$$k = \frac{[R_0] - [R]}{t}$$

(b) Write:

(i) Arrhenius equation.

(ii) The formula to calculate half-life period of zero order reaction.

Sol: (i) $k = Ae\frac{-Ea}{RT}$

(ii) $t_{1/2} = \frac{[R_0]}{2k}$

31. (a) Give any two differences between lyophilic and lyophobic collides.

Sol:

S.No.	Property	Lyophilic colloids	Lyophobic colloids
1.	Affinity	There is an affinity between dispersed phase and dispersion medium.	There is no affinity between dispersed phase and dispersion medium.
2.	Preparation	Easily prepared by direct mixing of dispersed phase and dispersion medium.	Preparation is difficult and special techniques are required.

(b) Write the two steps involved in the mechanism of enzyme catalyzed reaction.

Sol: Step 1: Binding of enzyme to substrate to form an activated complex

$E+S \rightarrow ES^{+}$

Step 2: Decomposition of the activated complex to form product.

$ES^{+} \rightarrow E+P$

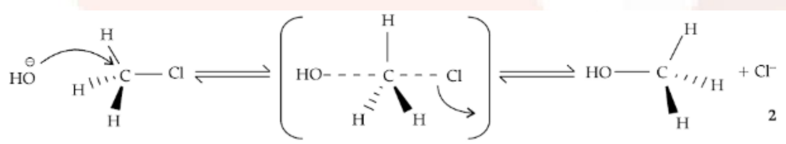
(c) What is the entropy change (ΔS) for adsorption?

Sol: Decreases or $\Delta S = -ve$ or $\Delta S < 0$.

V. Answer any Four of the following. Each question carries 5 marks.

32. (a) Write S_N^2 mechanism for conversion of methyl chloride to methyl alcohol.

Sol: (a) S_N^1 mechanism follows 2nd order kinetics and the mechanism is only on step.



(b) Aryl halides are extremely less reactive towards nucleophilic substitution reactions. Give any two reasons.

Sol: (i) C-X bond acquires double bond character due to resonance.

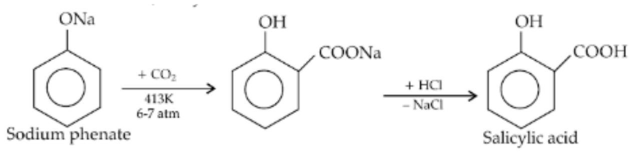
(ii) Difference in hybridization of carbon atom in C-X bond: In haloarene the C-atom attached to halgen is sp^2 hybridized that is grater's character is more electromotive and can hold the electron pair of C-X more tightly.

(c) What is asymmetric carbon?

Sol: Carbon atom is bonded to four different atoms or groups.

33. (a) Explain Kolbe's reaction with equation.

Sol: When sodium phenate is heated with carbon dioxide at 413K under 6-7 atm pressure followed by acidification with HCl, salicylic acid is formed.



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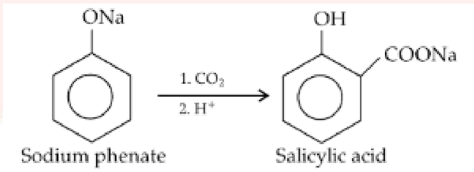
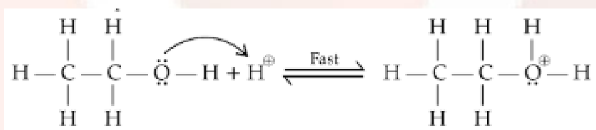
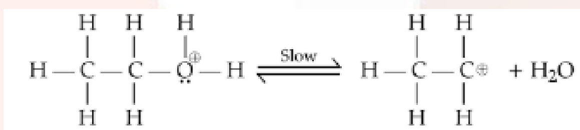


(b) Write the three steps involved in the mechanism of acid catalyzed dehydration of ethanol to ethene.

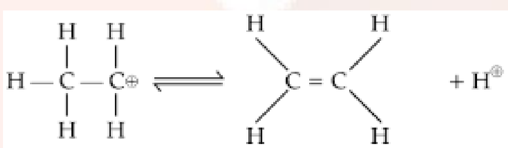
Sol. Step 1: Formation of protonated alohol:



Step 2: Formation of carbonation:

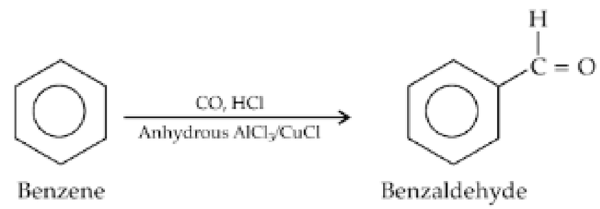


Step 3: Formation of proton:



34. (a) How benzene is converted into benzaldehyde by Gatterman-Koch reaction? Write equation.

Sol: When benzene is treated with carbon monoxide and HCl in the presence of Anhydrous $AlCl_3$, benzaldehyde is formed.



(b) Complete and name the following reation.

$$>C=O \xrightarrow[HCl]{Zn-Hg}$$

Sol:

$$>CH_2 \quad \text{OR} \quad CH_3-CH_2-CH_3$$

Clemmensen reduction

(c) What is the effect of electron withdrawing group on the acidity carboxylic acid?

Sol: Increases.

35. (a) How primary amine is prepared by Hoffmann bromamide degradation reaction? Write equation.

Sol:

(a) When amide is heated with bromine and sodium hydroxide, 1^0 –amine is formed.

$$R-CO-NH_2 + Br_2 + 4NaOH(aq) \xrightarrow{\Delta} R-NH_2 + Na_2CO_3 + 2NaBr + 2H_2O$$

(b) (i) Write IUPAC name of $CH_3CH_2NH_2$.

(ii) Arrange the following amines in the order of their increasing basic strength in aqueous solution. $(CH_3)_3N$, $(CH_3)_2NH$, CH_3NH_2.

Sol: (i) Ethanamine

(ii) $(CH_3)_3N < CH_3NH_2 < (CH_3)_2NH$

(c) Complete the following equation.

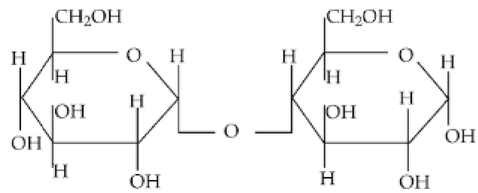
$$C_6H_5NH_2 + NaNO_2 + 2HCl \xrightarrow{273-278K}$$

Sol:

$$C_6H_5NH_2 + NaNO_2 + 2HCl \xrightarrow{273-278K} C_6H_5N_2^+Cl^- + Nacl + H_2O$$

36. (a) Write the Howorth structure of maltose.

Sol:



(b) Give an example for

(i) Globular proteins.

(ii) Naturally occurring optically inactive amino acid.

Sol: (i) Hoemoglobin or Albumin or Insulin or thyroglobumin or fibrinogen or venoms (of snakes, scorpion, wasp, bees).

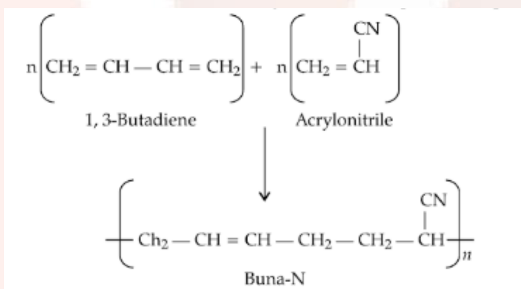
(ii) Glycine.

(c) Name the nucleic acid which is responsible for genetic information.

Sol: DNA (or) Deoxyribo nucleic acid.

37. (a) Explain the preparation of Buna- N with equation.

Sol: By co- polymerization of 1,3- butadiene and acrylo nitrile in the presence of peroxide or sodium catalyst.



(b) Name the monomer present in the following polymer

(i) Poly vinyl chloride.

(ii) Natural rubber.

Sol: (i) Vinyl chloride

(ii) Isoprene or 2-methyl-1, 3-butadiene.

(c) Give an example of bio- degradable polymer.

Sol:

Poly $\beta - hydroxybutyrate - co - \beta - hydroxy$ valerate (PHVB) or Nylon 2-Nylon 6 or polyglycolic acid (PGA) or polylactic acid (PLA) or polycaprolactone (PCL) or any natural polymer.